

Analysis and Evaluation of Daily PM₁₀ Forecasting in Jakarta Using a Hybrid Random Forest Regressive and ARIMA Model Based on Feature Engineering

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Abstract—Air quality reporting in Jakarta is retrospective, which limits proactive environmental mitigation. Following the Design Science Research Methodology (DSRM), this study analyzes and evaluates a sequential hybrid architecture that combines a Random Forest (RF) regressor with an ARIMA model to forecast next-day ($t+1$) PM₁₀ levels. Using 5,538 daily Air Pollutant Standard Index (ISPU) records from Jakarta (2010–2025), an 18-feature matrix is constructed strictly from information available up to the previous day, comprising PM₁₀ lags, rolling statistics, lagged gaseous pollutants, a CO $\times$ O₃ photochemical proxy, and calendar indicators. RF hyperparameters are tuned with the Tree-structured Parzen Estimator in Optuna under forward-chaining validation on the training period only, and an ARIMA model fitted to out-of-bag RF residuals is applied as a rolling one-step correction. On a chronologically held-out test period of 1,043 days, the hybrid model achieves an RMSE of 14.73, an MAE of 10.34, and an R^2 of 0.37, reducing RMSE by 17.0% relative to a persistence baseline (Diebold–Mariano $p < 0.001$). The ARIMA residual correction reduces RMSE by only 0.63% relative to the tuned RF, a difference that is not statistically significant ($p = 0.17$), indicating that predictive skill stems mainly from temporal feature engineering. TreeSHAP analysis shows that short-term PM₁₀ persistence and seasonality dominate the model’s decisions.

Index Terms—air quality forecasting, particulate matter, hybrid model, Random Forest, ARIMA, feature engineering, DSRM, explainable AI, TreeSHAP

I. INTRODUCTION

Urban centers in developing countries face persistent challenges related to air quality degradation. Jakarta, the economic center of Indonesia, frequently records particulate matter levels above the guideline values of the World Health Organization (WHO). Particulate matter with an aerodynamic diameter below 10 μm (PM₁₀) is of particular concern because it can penetrate the upper respiratory tract and is associated with long-term risks of Chronic Obstructive Pulmonary Disease (COPD) and cardiovascular disease.

The Indonesian government communicates air quality through the Air Pollutant Standard Index (ISPU) defined in Ministry of Environment and Forestry Regulation No. P.14/2020 [1]. The published index summarizes conditions over the preceding 24 hours and is therefore inherently retrospective, as summarized in Fig. 1. Moving from reactive

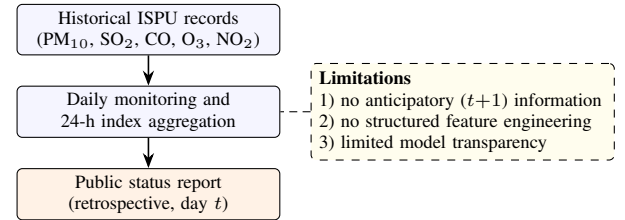


Fig. 1. Workflow of the existing retrospective air quality reporting system in Jakarta and its limitations.

reporting to next-day ($t+1$) forecasting would allow public authorities to issue warnings before high-pollution days occur.

Statistical and machine learning models have been widely applied to air quality prediction [2]. Tree ensembles such as Random Forest (RF) [3] capture nonlinear interactions among pollutants but treat samples as independent and therefore do not model temporal autocorrelation explicitly. Conversely, AutoRegressive Integrated Moving Average (ARIMA) models capture linear temporal dependence but cannot represent nonlinear cross-pollutant effects [4]–[6]. Hybrid schemes following Zhang’s decomposition [7] combine both: a nonlinear learner models the signal and ARIMA models its residuals. Existing RF–ARIMA studies, however, often feed raw pollutant values directly and seldom document how temporal features are constructed or how information leakage is avoided [8], and black-box models complicate validation by environmental analysts [9].

This study addresses three research questions:

- RQ1** To what extent does a sequential RF–ARIMA hybrid improve next-day PM₁₀ forecasting accuracy over single-model and naive baselines?
- RQ2** How do the individual groups of engineered temporal and chemical features affect forecasting error?
- RQ3** Do post-hoc TreeSHAP explanations reveal decision behavior consistent with atmospheric processes?

The contributions of this study are as follows:

- A leakage-free next-day forecasting pipeline in which all 18 features use only information available up to day t –

- 1, with past-only imputation and evaluation restricted to days with measured PM_{10} .
- A sequential RF–ARIMA hybrid that fits ARIMA to out-of-bag RF residuals and applies it as a rolling one-step correction.
- A rigorous evaluation on 15 years of Jakarta ISPU data against persistence, rolling ARIMA, and XGBoost baselines, with an ablation study and Diebold–Mariano significance tests.
- A TreeSHAP analysis of global and feature-level attributions, including the $\text{CO} \times \text{O}_3$ proxy.

II. METHODOLOGY

The study follows the Design Science Research Methodology (DSRM) [10], [11] in five phases: (1) problem identification through analysis of the existing reporting workflow, (2) objective definition based on the ISPU standard, (3) design and development of the feature matrix and hybrid pipeline, (4) demonstration on the Jakarta dataset, and (5) evaluation through benchmarking, ablation, and explainability analysis.

A. Pre-Development Analysis via AHP

Before implementation, the Analytical Hierarchy Process (AHP) [12] was used to structure the choice of modeling paradigm against five criteria: Interpretability (INT), Accuracy (ACC), Pattern Complexity (PCX), Chemical Logic Capability (CHM), and Ease of Implementation (IMP). The pairwise comparison matrix produced weights of 0.352 (INT), 0.295 (ACC), 0.165 (PCX), 0.097 (CHM), and 0.091 (IMP), with a Consistency Ratio of 0.037 (< 0.1). The hybrid alternative obtained a global priority of 36.5%, nearly tied with standalone ARIMA (36.4%) and ahead of standalone RF (27.1%). Because the top two alternatives are essentially tied, the AHP result is used only as a design rationale, and the choice is evaluated empirically in Section III.

B. Dataset and Preprocessing

The data are the public daily ISPU records for DKI Jakarta [13], covering January 1, 2010 to February 28, 2025 (5,538 days). Each day is reported by one of five monitoring stations (DKI1–DKI5; 531, 1,137, 1,116, 1,674, and 1,079 days, respectively) and contains ISPU sub-index values for PM_{10} , SO_2 , CO , O_3 , and NO_2 . All values in this paper are therefore dimensionless ISPU sub-indices rather than mass concentrations. $\text{PM}_{2.5}$ was excluded because 72.6% of its values are missing. PM_{10} is missing on 5.7% of days (315 days).

To avoid look-ahead bias, missing values are filled by forward-filling (using only past observations), and the filled values are used only to construct features. Days on which PM_{10} was not measured are never used as targets in either training or evaluation. After discarding the first week needed for lag construction, 5,215 samples remain. They are split chronologically into a training set of 4,172 days (Jan. 8, 2010–Oct. 22, 2021) and a test set of 1,043 days (Oct. 23, 2021–Feb. 28, 2025). The test set has a mean of 55.78 and a standard deviation of 18.53.

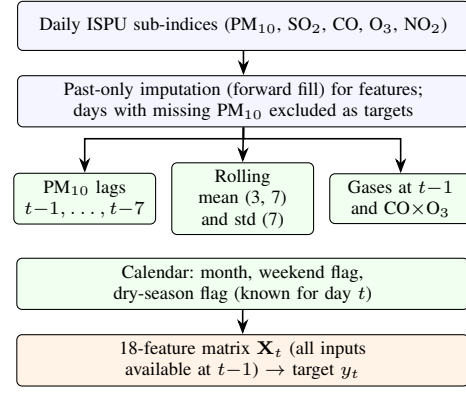


Fig. 2. Feature engineering pipeline. Every feature uses only information available before the forecast day.

C. Feature Engineering

For a target y_t (the PM_{10} sub-index on day t), 18 features are computed using information available up to day $t - 1$ (Fig. 2):

- 1) *Temporal lags*: $y_{t-1}, \dots, y_{t-7}$ (pm10_lag_1–pm10_lag_7), capturing short-term persistence and the weekly activity cycle.
- 2) *Rolling statistics*: 3- and 7-day means (pm10_ma_3, pm10_ma_7) and the 7-day standard deviation (pm10_std_7) of $y_{t-1}, \dots$, which smooth noise and quantify recent volatility.
- 3) *Lagged gaseous pollutants*: SO_2 , CO , O_3 , and NO_2 on day $t - 1$.
- 4) *Photochemical proxy*: $\text{CO}_{t-1} \times \text{O}_{3,t-1}$ (co_x_o3_lag_1). CO acts as a tracer of combustion emissions and O_3 as an indicator of photochemical oxidation, so their product serves as a proxy for conditions that favor secondary particle formation when no solar radiation data are available.
- 5) *Calendar indicators*: month, a weekend flag, and a season flag (dry season, April–October), which are known in advance for day t .

D. Sequential Hybrid Model

Following Zhang [7], the series is treated as the sum of a nonlinear component N_t and a linearly autocorrelated residual component L_t :

$$y_t = N_t + L_t. \quad (1)$$

In the first stage, an RF regressor estimates the nonlinear component by averaging B regression trees:

$$\hat{N}_t = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{X}_t). \quad (2)$$

Because in-sample RF residuals are strongly shrunk by overfitting, the training residuals are computed from out-of-bag (OOB) predictions $\hat{N}_t^{\text{OOB}}$:

$$e_t = y_t - \hat{N}_t^{\text{OOB}}. \quad (3)$$

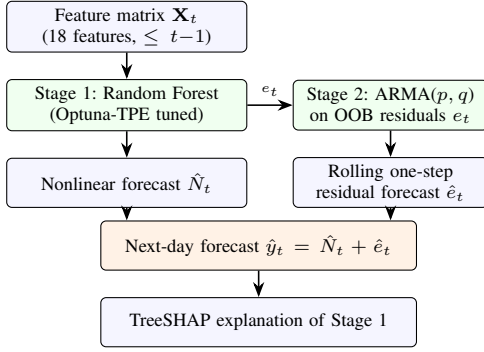


Fig. 3. Sequential hybrid architecture: RF models nonlinear structure and ARMA corrects residual autocorrelation.

In the second stage, an $\text{ARMA}(p, q)$ model (ARIMA with $d = 0$, since the residuals are stationary) is fitted to e_t :

$$\hat{e}_t = c + \sum_{i=1}^p \phi_i e_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j}, \quad (4)$$

with $(p, q) \in \{0, \dots, 5\}^2$ selected by the Akaike Information Criterion on the training residuals. During testing, the residual of day $t - 1$ is known once y_{t-1} is observed, so the ARMA model produces a rolling one-step forecast without re-estimating its parameters. The hybrid forecast is

$$\hat{y}_t = \hat{N}_t + \hat{e}_t. \quad (5)$$

The architecture is shown in Fig. 3.

E. Hyperparameter Optimization and Validation

RF hyperparameters are optimized with the Tree-structured Parzen Estimator (TPE) [14] implemented in Optuna [15]. Each of 30 trials is scored by the mean RMSE of a 5-fold forward-chaining *TimeSeriesSplit* on the training set only, so the test set is never used for model selection. The search space is $n_{\text{estimators}} \in [100, 300]$, $\text{max_depth} \in [5, 30]$, $\text{min_samples_leaf} \in [1, 10]$, and $\text{max_features} \in \{\sqrt{\cdot}, \log_2, \text{all}\}$.

F. Baselines and Evaluation

The hybrid is compared with (i) persistence ($\hat{y}_t = y_{t-1}$), (ii) a univariate ARIMA(5,1,0) on PM_{10} applied as a rolling one-step forecast, (iii) XGBoost on the same 18 features with fixed, untuned settings (300 trees, learning rate 0.05, depth 6), and (iv) the tuned RF alone. Accuracy is measured by RMSE, MAE, and R^2 . Differences in squared error are tested with the Diebold–Mariano (DM) test [16] with the small-sample correction of Harvey et al. [17]. All experiments are implemented in Python with scikit-learn, statsmodels, XGBoost, Optuna, and SHAP.

III. RESULTS AND DISCUSSION

A. Stationarity

The Augmented Dickey–Fuller (ADF) test [18] rejects a unit root for the PM_{10} series (statistic -6.52 , $p < 0.001$),

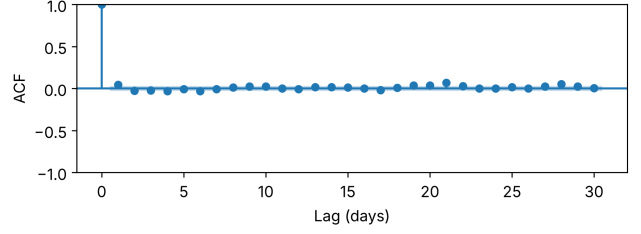


Fig. 4. Autocorrelation function of out-of-bag RF residuals on the training set (shaded band: 95% confidence interval).

TABLE I
NEXT-DAY FORECASTING PERFORMANCE ON THE TEST SET (1,043 DAYS)

Model	RMSE	MAE	R^2	DM [†]	p
Persistence	17.75	11.96	0.08	3.75	<0.001
ARIMA(5,1,0), rolling	15.23	10.70	0.32	2.31	0.021
XGBoost (untuned)	16.52	12.09	0.20	7.69	<0.001
RF (Optuna-tuned)	14.82	10.55	0.36	1.38	0.168
Hybrid RF–ARMA	14.73	10.34	0.37	–	–

[†]DM statistic of each model versus the hybrid (squared-error loss); a positive value means the hybrid is more accurate.

and the KPSS test [19] does not reject stationarity at the 5% level (statistic 0.435, $p = 0.062$). The OOB residuals of the RF retain weak but statistically significant autocorrelation (Ljung–Box $Q(7) = 21.93$, $p = 0.003$; $Q(14) = 29.98$, $p = 0.008$), although all autocorrelation coefficients are small (Fig. 4). The AIC selected ARMA(5,4) for the residual stage.

B. Hyperparameter Optimization

The TPE search converged to $n_{\text{estimators}} = 200$, $\text{max_depth} = 9$, $\text{min_samples_leaf} = 10$, and $\text{max_features} = \log_2$, with a mean cross-validation RMSE of 16.99. The preference for shallow trees and large leaves indicates strong regularization, which is consistent with the high day-to-day noise of the target. Training the RF and fitting the residual model, including the ARMA order search, takes about 40 s on a 2-vCPU machine, and inference takes less than 0.1 ms per day, which is negligible for a daily operational update.

C. Benchmark Comparison (RQ1)

Table I summarizes test performance. The hybrid achieves the lowest error (RMSE 14.73, MAE 10.34, $R^2 = 0.37$). Relative to persistence, it reduces RMSE by 17.0% and MAE by 13.5%, and relative to the rolling univariate ARIMA it reduces RMSE by 3.3%; these differences, as well as the difference with XGBoost, are significant at the 5% level. XGBoost with the same features but without tuning performs worse than the tuned RF, which underlines the importance of regularization on this noisy target.

The contribution of the ARIMA stage itself is small. Compared with the tuned RF alone, the hybrid lowers RMSE by 0.63% and MAE by 1.97%, and the DM test does not find this difference significant ($p = 0.168$). This is consistent with Fig. 4: once lagged and rolling PM_{10} features are included,

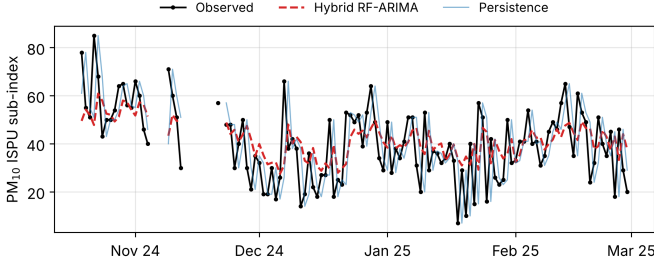


Fig. 5. Observed PM_{10} sub-index and next-day forecasts over the last 120 days of the test period. Gaps correspond to days without a PM_{10} measurement.

TABLE II
ABLATION OF FEATURE GROUPS (RF WITH 100 TREES + ARMA
RESIDUAL STAGE)

Configuration	#Feat.	RMSE	MAE	R^2
A: lagged gaseous pollutants	4	16.50	12.18	0.21
B: A + PM_{10} lags & rolling stats	14	15.24	10.91	0.32
C: B + $CO \times O_3$ & calendar (full)	18	15.10	10.82	0.33

the RF already absorbs most of the temporal dependence, leaving little linear structure for the residual model. Regarding RQ1, the hybrid is therefore significantly better than naive and univariate baselines, but its advantage over a well-regularized RF with temporal features is marginal. Fig. 5 shows that the forecasts follow the level and slower variations of the series but smooth out abrupt one-day spikes.

D. Ablation Study (RQ2)

Table II isolates the feature groups using an untuned RF (100 trees) with the same residual stage. Adding PM_{10} lags and rolling statistics produces the largest gain (RMSE 16.50 to 15.24, -7.6%), while the interaction proxy and calendar indicators add a further, smaller improvement (to 15.10, -0.9%). Comparing configuration C with the tuned model in Table I (RMSE 14.73) shows that hyperparameter optimization contributes an additional 2.5% reduction.

E. Error by Reporting Station and Extreme Events

Table III groups test errors by the station that reported each day. DKI4 (Lubang Buaya), which reports the highest mean levels and half of the test days, has the largest RMSE (15.90), whereas DKI3 (Jagakarsa) has the lowest (11.41). The model performs poorly on rare extreme events: on the 8 test days with a sub-index of at least 101 (“Unhealthy”), the RMSE rises to 85.57, because such spikes are rarely preceded by elevated values on the previous days. Over the whole test set, the mean bias is small (-1.22).

F. Explainability with TreeSHAP (RQ3)

TreeSHAP [20]–[22] was applied to the RF stage on 400 randomly sampled test days (Fig. 6). The three largest mean absolute attributions belong to the 3-day mean (2.58), the 7-day mean (2.49), and the 1-day lag of PM_{10} (2.48), confirming that short-term persistence of particulate matter in the lower

TABLE III
HYBRID TEST ERROR GROUPED BY REPORTING STATION

Station	Days	Mean y	RMSE	MAE
DKI1 (Bunderan HI)	64	42.55	12.87	9.93
DKI2 (Kelapa Gading)	136	52.65	13.86	11.33
DKI3 (Jagakarsa)	168	45.70	11.41	8.64
DKI4 (Lubang Buaya)	524	63.62	15.90	10.32
DKI5 (Kebon Jeruk)	151	48.24	15.28	11.60

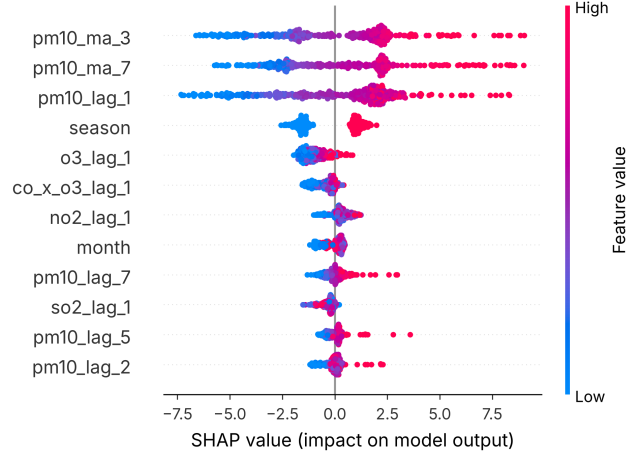


Fig. 6. TreeSHAP summary of the 12 most influential features (400 test days). Color encodes the feature value.

atmosphere dominates next-day forecasts. The dry-season flag ranks fourth (1.35): dry-season days receive positive contributions, consistent with reduced wet deposition. Among chemical inputs, lagged O_3 (1.06) and the $CO \times O_3$ proxy (0.52) rank highest. The SHAP value of the proxy increases monotonically with its magnitude (Spearman $\rho = 0.83$), and its mean contribution is higher in the dry season (-0.19) than in the wet season (-0.81). This pattern is consistent with the intended interpretation as a photochemical indicator, although its overall effect is modest.

G. Limitations

The dataset contains one reporting station per day, so the series mixes spatial and temporal variation. Meteorological drivers such as wind, rainfall, and boundary-layer height are not available, which limits the explained variance ($R^2 = 0.37$) and the ability to anticipate extreme events.

IV. CONCLUSION

This study analyzed and evaluated a sequential RF–ARIMA hybrid for next-day PM_{10} forecasting in Jakarta under a strictly leakage-free design. For RQ1, the hybrid achieved the best test performance (RMSE 14.73, $R^2 = 0.37$) and significantly outperformed persistence (-17.0% RMSE) and a rolling univariate ARIMA (-3.3%), but the ARIMA residual stage improved on the tuned RF by only 0.63% , which is not statistically significant. For RQ2, PM_{10} lags and rolling statistics provided most of the gain, while the interaction and

calendar features contributed a smaller improvement. For RQ3, TreeSHAP showed that recent PM₁₀ levels and the dry season dominate the forecasts and that the CO × O₃ proxy behaves consistently with its photochemical interpretation. In practice, a well-regularized RF with temporal features captures most of the achievable skill from pollutant data alone. Future work will incorporate meteorological variables, station-specific models, and approaches targeted at extreme pollution events.

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